

DATA 335 - STATISTICAL MODELLING

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1. THE ONE-SAMPLE t -TEST

Suppose we want to test

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

using a random sample

$$X_0, \dots, X_{n-1} \sim N(\mu, \sigma^2).$$

Let $\bar{X}$ and s^2 be the sample mean and sample variance, respectively:

$$\bar{X} = \frac{1}{n} \sum_{i < n} X_i, \quad S^2 = \frac{1}{n-1} \sum_{i < n} (X_i - \bar{X})^2$$

We form the *test statistic*

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S}.$$

Write

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)/\sigma}{S/\sigma}.$$

$$\sqrt{n}(\bar{X} - \mu_0)/\sigma \sim N(0, 1)$$

$$\frac{S}{\sigma} = \sqrt{\frac{S^2}{\sigma^2}} = \sqrt{\frac{1}{n-1} \sum_{i < n} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2} = \sqrt{V/(n-1)},$$

where

$$V = \sum_{i < n} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi_{n-1}^2$$

Under the null hypothesis,

$$T \sim t_{n-1}.$$

Under H_0 ,

$$\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right),$$

so

$$U := \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \sim N(0, 1).$$

$$V := (n-1) \frac{S^2}{\sigma^2} = \sum_{i < n} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi_{n-1}^2$$

Thus,

$$T = \frac{U}{\sqrt{V/(n-1)}} \sim t_{n-1}.$$

2. THE TWO-SAMPLE t -TEST

$$\mathbb{V}(\bar{X} - \bar{Y}) = \mathbb{V}(X) + \mathbb{V}(Y) = \frac{\sigma^2}{m} + \frac{\sigma^2}{n} = \sigma^2 \left(\frac{1}{m} + \frac{1}{n} \right)$$

$$S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2} = \frac{1}{m+n-2} \sum_{i < m} (X_i - \bar{X})^2 + \sum_{j < n} (Y_j - \bar{Y})^2$$

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{m} + \frac{1}{n}}} = \frac{\frac{\bar{X} - \bar{Y}}{\sigma \sqrt{\frac{1}{m} + \frac{1}{n}}}}{\frac{S_p}{\sigma}}$$

$$\frac{\bar{X} - \bar{Y}}{\sigma \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim N(0, 1)$$

$$\frac{S_p}{\sigma} = \sqrt{\frac{S_p^2}{\sigma^2}} = \sqrt{\frac{1}{m+n-2} \left(\sum_{i < m} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + \sum_{j < n} \left(\frac{Y_j - \bar{Y}}{\sigma} \right)^2 \right)} = \sqrt{\frac{V}{m+n-2}},$$

where

$$V = \sum_{i < m} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + \sum_{j < n} \left(\frac{Y_j - \bar{Y}}{\sigma} \right)^2 \sim \chi_{m+n-2}^2$$

Therefore,

$$T \sim t_{m+n-2}.$$

If $m = n$, then

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{2}{n}}} \sim t_{2n-2}.$$

3. ONE-WAY ANOVA

$$\bar{Y}_i = \frac{1}{J} \sum_{j < J} Y_{ij}$$

$$\bar{Y} = \frac{1}{IJ} \sum_{i < I} \sum_{j < J} y_{ij} = \frac{1}{I} \sum_{i < I} \bar{Y}_i$$

When $I = 2$,

$$\bar{Y} = \frac{\bar{Y}_1 + \bar{Y}_2}{2}.$$

Therefore,

$$\bar{Y}_1 - \bar{Y} = \frac{Y_1 - Y_2}{2}, \quad \bar{Y}_2 - \bar{Y} = \frac{Y_2 - Y_1}{2},$$

Therefore,

$$(\bar{Y}_1 - \bar{Y})^2 + (\bar{Y}_2 - \bar{Y})^2 = \frac{1}{2} (\bar{Y}_1 - \bar{Y}_2)^2$$

$$\begin{aligned} \text{SSB} &= J ((\bar{Y}_1 - \bar{Y})^2 + (\bar{Y}_2 - \bar{Y})^2) \\ &= \frac{J}{2} (\bar{Y}_1 - \bar{Y}_2)^2 \end{aligned}$$

$$S_p^2 = \frac{1}{I(J-1)} \left(\sum_{j < J} (Y_{1j} - \bar{Y}_1)^2 \right)$$

4. HIERARCHICAL MODELS

$$\begin{aligned} x_i &\sim N(\mu, \sigma^2) & i < I \\ y_{ij} \mid x_i &\sim N(x_i, \tau^2) & i < I, j < J \end{aligned}$$

Law of total expectation:

$$\begin{aligned} \mathbb{E}(y_{ij}) &= \mathbb{E}(\mathbb{E}(y_{ij} \mid x_i)) \\ &= \mathbb{E}(x_i) \\ &= \mu \end{aligned}$$

Law of total variance:

$$\begin{aligned}\mathbb{V}(y_{ij}) &= \mathbb{E}(\mathbb{V}(y_{ij} \mid x_i)) + \mathbb{V}(\mathbb{E}(y_{ij} \mid x_i)) \\ &= \mathbb{E}(\tau^2) + \mathbb{V}(x_i) \\ &= \tau^2 + \sigma^2\end{aligned}$$

Therefore

$$y_{ij} \sim N(\mu, \sigma^2 + \tau^2).$$

$$Q(\theta_t, \theta) = \mathbb{E}_{x \sim f(x|y, \theta_t)} (\log f(x, y \mid \theta))$$

$$\begin{aligned}f(x \mid y) &\propto f(y \mid x)f(x) \\ &\propto \exp\left(-\frac{1}{2\tau^2}(y-x)^2\right) \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)\end{aligned}$$

$$\begin{aligned}\frac{1}{\tau^2}(y-x)^2 + \frac{1}{\sigma^2}(x-\mu)^2 &= \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2}\right)x^2 - 2\left(\frac{y}{\tau^2} + \frac{\mu}{\sigma^2}\right)x + \frac{y^2}{\tau^2} + \frac{\mu^2}{\sigma^2} \\ &= Ax^2 - 2B(y)x + C(y) \\ &= A\left(x^2 - 2\frac{B(y)}{A}x + \frac{B(y)^2}{A^2}\right) + D(y) \\ &= A\left(x - \frac{B(y)}{A}\right)^2 + D(y)\end{aligned}$$

$$f(x \mid y) \propto \exp ()$$

5. RIGHT-CENSORED DATA

$$X_i \sim N(\mu, \sigma^2), \quad Y_i = \min(X_i, a)$$

Given estimates μ_t, σ_t^2 of μ and σ^2 .

$$\mathbb{E}_{x_i \sim p_t(x_i|y_i)} [\log f(x_i)]$$

$$\ell = \sum_i \left(\frac{1}{2} \log(2\pi) + \frac{1}{2} \log \sigma^2 + \frac{1}{2\sigma^2} (\mathbb{E}_t[x_i^2 \mid y_i] - 2\mu \mathbb{E}_t[x_i \mid y_i] + \mu^2) \right)$$

$$\frac{\partial \ell}{\partial \mu} = \sum_i \frac{1}{2\sigma^2} (-2 \mathbb{E}_t[x_i | y_i] + 2\mu) = \frac{1}{\sigma^2} \left(n\mu - \sum_i \mathbb{E}_t[x_i | y_i] \right) = 0$$

$$\mu = \frac{1}{n} \sum_i \mathbb{E}_t[x_i | y_i]$$

$$\frac{\partial \ell}{\partial \sigma^2} = \frac{n}{2\sigma^2} - \frac{1}{(2\sigma^2)^2} \sum_i \mathbb{E}_t[(x_i - \mu)^2 | y_i] = 0$$

$$\sigma^2 = \frac{1}{n} \sum_i \mathbb{E}_t[(x_i - \mu)^2 | y_i]$$

Let

$$I = \{i : y_i < b\}, \quad J = \{i : y_i = a\}.$$

If $i \in I$, then $\mathbb{E}[x_i | y_i] = y_i$.

If $i \in J$, then $\mathbb{E}[x_i | y_i] = \mathbb{E}_{x \sim N(\mu_t, \sigma_t^2)}[x | x \geq a]$.

Suppose $X \sim N(0, 1)$.

$$E_{X \sim N(0,1)}[X | X \geq a] = \frac{\varphi(a)}{1 - \Phi(a)} = m_1(a)$$

Suppose $X \sim N(\mu, \sigma^2)$.

$$E[X | X \geq a] = \sigma m_1(a) + \mu$$

6. TRUNCATED NORMAL DISTRIBUTIONS

Let $\Phi(x | \mu, \sigma)$ and $\varphi(x | \mu, \sigma)$ be the distribution and density functions, respectively, of the normal distribution with mean μ and standard deviation σ . As shorthand, write $\Phi(x)$ and $\varphi(x)$ for $\Phi(x | 0, 1)$ and $\varphi(x | 0, 1)$, respectively. In particular,

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

For parameters a, b, μ , and σ with $a < b$ and $\sigma > 0$, let $\text{TruncNorm}(a, b; \mu, \sigma)$ be the distribution with density

$$\varphi(x | a, b; \mu, \sigma) = \frac{\mathbf{1}_{[a,b]}(x) \varphi(x | \mu, \sigma)}{\Phi(b | \mu, \sigma) - \Phi(a | \mu, \sigma)}.$$

$$\frac{d\varphi}{dx} = -x\varphi(x)$$

Suppose $Z \sim N(0, 1)$.

$$\begin{aligned}\mathbb{E}[Z \mid a < Z < b] &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b x\varphi(x)dx \\ &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b \frac{d}{dx}(-\varphi(x))dx \\ &= \frac{\varphi(a) - \varphi(b)}{\Phi(b) - \Phi(a)}\end{aligned}$$

Special cases:

$$\begin{aligned}\mathbb{E}[Z \mid a < Z] &= \frac{\varphi(a)}{1 - \Phi(a)} = \psi(a) \\ \mathbb{E}[Z \mid Z < b] &= \frac{-\varphi(b)}{\Phi(b)}\end{aligned}$$

We have:

$$\begin{aligned}\frac{d}{dx}x\varphi(x) &= \varphi(x) + x\varphi'(x) \\ &= \frac{d}{dx}\Phi(x) - x^2\varphi(x)\end{aligned}$$

Rearrange:

$$\frac{d}{dx}(\Phi(x) - x\varphi(x)) = x^2\varphi(x).$$

Therefore:

$$\begin{aligned}\mathbb{E}[Z^2 \mid a < Z < b] &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b x^2\varphi(x)dx \\ &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b \frac{d}{dx}(\Phi(x) - x\varphi(x)) dx \\ &= \frac{1}{\Phi(b) - \Phi(a)} (\Phi(b) - b\varphi(b) - \Phi(a) + a\varphi(a)) \\ &= 1 + \frac{a\varphi(a) - b\varphi(b)}{\Phi(b) - \Phi(a)}\end{aligned}$$

Special cases:

$$\begin{aligned}\mathbb{E}[Z^2 \mid a < Z] &= 1 + \frac{a\varphi(a)}{1 - \Phi(a)} = 1 + a\psi(a) \\ \mathbb{E}[Z^2 \mid Z < b] &= 1 - \frac{b\varphi(b)}{\Phi(b)}\end{aligned}$$

Suppose $X \sim N(\mu, \sigma^2)$. Set

$$Z = \frac{X - \mu}{\sigma}, \quad \alpha = \frac{a - \mu}{\sigma}, \quad \beta = \frac{b - \mu}{\sigma}.$$

$$\begin{aligned}\mathbb{E}[X \mid a < X < b] &= \mu + \sigma \mathbb{E}[Z \mid \alpha < Z < \beta] \\ &= \mu + \sigma m_1(\alpha, \beta)\end{aligned}$$

Define

$$\psi(x) = \frac{\varphi(x)}{1 - \Phi(x)}$$

$$\mathbb{E}[X \mid a < X] = \mu + \sigma \psi(\alpha)$$

$$\begin{aligned}\mathbb{E}[X^2 \mid a < X < b] &= \mathbb{E}[(\mu + \sigma Z)^2 \mid \alpha < Z < \beta] \\ &= \mu^2 + 2\mu\sigma m_1(\alpha, \beta) + \sigma^2 m_2(\alpha, \beta)\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X^2 \mid a < X] &= \mu^2 + 2\mu\sigma\psi(\alpha) + \sigma^2(1 + \alpha\psi(\alpha)) \\ &= \mu^2 + \sigma^2 + \sigma\psi(\alpha)(2\mu + \sigma\alpha)\end{aligned}$$

$$\begin{aligned}u_{t,i} &= \begin{cases} x_i & \text{if } x_i < a, \\ \mu_t + \sigma_t \psi(\alpha_t) & \text{otherwise} \end{cases} \\ v_{t,i} &= \begin{cases} x_i^2 & \text{if } x_i < a \\ \mu_t^2 + \sigma_t^2 + \sigma_t \psi(\alpha_t)(2\mu_t + \sigma_t \alpha_t) & \text{otherwise} \end{cases}\end{aligned}$$

$$\begin{aligned}\mu_{t+1} &= \frac{1}{n} \sum_{i < n} u_{t,i} \\ \sigma_{t+1}^2 &= \frac{1}{n} \sum_{i < n} (v_{t,i} - 2\mu_{t+1} u_{t,i} + \mu_{t+1}^2) \\ &= \frac{1}{n} \sum_{i < n} v_{t,i} - \mu_{t+1}^2\end{aligned}$$

7. EXPONENTIAL CENSORING

$$\varphi(x) = \lambda e^{-\lambda x}$$

Suppose $X \sim \text{Exp}(\lambda)$.

$$\begin{aligned}\mathbb{E}[X \mid X > a] &= \frac{1}{e^{-\lambda a}} \int_a^\infty x \lambda e^{-\lambda x} dx \\ &= \int_a^\infty \lambda x e^{-\lambda(x-a)} dx \\ &= \int_0^\infty \lambda(t+a) e^{-\lambda t} dt \\ &= \int_0^\infty t \varphi(t) dt + a \int_0^\infty \varphi(t) dt \\ &= \frac{1}{\lambda} + a\end{aligned}$$

8. HYPOTHESIS TESTING

Let $\Theta = \Theta_0 \sqcup \Theta_1$ and let R be a rejection region. Define the *power function* $\beta : \Theta \rightarrow \mathbb{R}$ by

$$\beta(\theta) = \mathbb{P}_\theta(R).$$

The *size* of the test is

$$\alpha = \sup_{\theta \in \Theta_0} \beta(\theta).$$

Suppose $X \sim \text{Binomial}(5, \theta)$

$$\Theta_0 = \{\theta \leq 1/2\}, \quad \Theta_1 = \{\theta > 1/2\},$$

Let $\{X = 5\}$ be the rejection region.

$$\beta(\theta) = \mathbb{P}_\theta(X = 5) = \binom{5}{5} \theta^5 (1-\theta)^{5-5} = \theta^5$$

The size of the test is

$$\alpha = \sum_{\theta \leq 1/2} \theta^5 = 1/32 = 0.03125.$$

Suppose $X \sim \text{Normal}(\theta, \sigma^2)$ with σ^2 known. Let

$$\Theta_0 = \{\theta \leq \theta_0\}, \quad \Theta_1 = \{\theta > \theta_0\},$$

Let $\{X > x\}$ be the rejection region. Then

$$\begin{aligned}\beta(\theta) = \mathbb{P}_\theta(X > x) &= \mathbb{P}_\theta\left(\frac{X - \theta}{\sigma} > \frac{x - \theta}{\sigma}\right) \\ &= \mathbb{P}\left(Z > \frac{x - \theta}{\sigma}\right)\end{aligned}$$

Since $\mathbb{P}\left(Z > \frac{x-\theta}{\sigma}\right)$ is an increasing function of θ ,

$$\sup_{\theta \leq \theta_0} \beta(\theta) = \mathbb{P}\left(Z > \frac{x - \theta_0}{\sigma}\right)$$

9. THE GERMAN TANK PROBLEM

Consider the problem of estimating θ from a random sample

$$X_0, \dots, X_{n-1} \sim \text{Uniform}[0, \theta].$$

Since

$$\mathbb{E}[\bar{X}] = \frac{\theta}{2},$$

it follows that

$$U = 2\bar{X}$$

is an unbiased estimator of θ .

Using the fact that

$$\mathbb{V}[X_i] = \frac{\theta^2}{12},$$

it follows that

$$\mathbb{V}[U] = \frac{\theta^2}{3n}.$$

Another estimator is

$$V = \max\{X_0, \dots, X_{n-1}\}.$$

This estimator, however, is biased downward:

$$\mathbb{E}[V] < \theta.$$

We can actually compute this bias exactly.

Write F_V for the cumulative distribution function of V :

$$F_V(x) = \mathbb{P}[V < x].$$

Since $V < x$ if and only if $X_i < x$ for all i and, moreover, the X_i are independent,

$$F_V(x) = \prod_{i < n} \mathbb{P}[X_i < x] = \frac{x^n}{\theta^n}$$

The density $f_V(x)$ of V is given by the derivative of $F_V(x)$:

$$f_V(x) = \frac{n}{\theta^n} x^{n-1}.$$

Therefore,

$$\begin{aligned}\mathbb{E}[V] &= \int_0^\theta x f_V(x) dx \\ &= \frac{n}{\theta^n} \int_0^\theta x^n dx \\ &= \frac{n}{\theta^n} \frac{\theta^{n+1}}{n+1} \\ &= \frac{n}{n+1} \theta\end{aligned}$$

It follows that

$$W = \frac{n+1}{n} V$$

is an unbiased estimator of θ .

For the German tank problem with

$$X_0 = 19, \quad X_1 = 40, \quad X_2 = 42, \quad X_3 = 60,$$

(see the Wikipedia article) this gives

$$W = 75.$$

While we're at it, let's compute the second moment of V :

$$\begin{aligned}\mathbb{E}[V^2] &= \int_0^\theta x^2 f_V(x) dx \\ &= \frac{n}{\theta^n} \int_0^\theta x^{n+1} dx \\ &= \frac{n}{\theta^n} \frac{\theta^{n+2}}{n+2} \\ &= \frac{n}{n+2} \theta^2\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{V}[V] &= \mathbb{E}[V^2] - \mathbb{E}[V]^2 \\ &= \frac{n}{n+2} \theta^2 - \left(\frac{n}{n+1} \theta \right)^2 \\ &= \frac{n}{(n+2)(n+1)^2} \theta^2\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{V}[W] &= \left(\frac{n+1}{n} \right)^2 \mathbb{V}[V] \\ &= \frac{\theta^2}{n(n+2)}\end{aligned}$$

Notice that

$$\mathbb{V}[W] = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{3n} = \mathbb{V}[U]$$

for all $n > 1$.

Thus, W is a better estimator than U whenever $n > 1$.

10. MORE GERMAN TANK PROBLEM

Now let's try to approach the German tank problem discretely.

We'll view a sample of k captured German tanks as a sample of size k drawn without replacement from $\{1, \dots, n\}$.

Write $\Omega_{n,k}$ for the set of k -tuples of distinct elements of $\{1, \dots, n\}$.

$$|\Omega_{n,k}| = k! \binom{n}{k}$$

Now consider

$$\Omega_{n,k,m} := \{(s_1, \dots, s_k) \in \Omega_{n,k} : \max\{s_1, \dots, s_k\} = m\}$$

To make an element of $\Omega_{n,k,m}$, draw an element of $\Omega_{m-1,k-1}$ and then insert m somewhere in the sequence. Thus,

$$|\Omega_{n,k,m}| = k! \binom{m-1}{k-1}$$

Therefore,

$$p(m | n) = \frac{\binom{m-1}{k-1}}{\binom{n}{k}} \mathbf{1}_{[m_s, \infty)}(n) \propto \frac{\mathbf{1}_{[m, \infty)}(n)}{\binom{n}{k}}.$$

$$p(s | n) = \frac{\mathbf{1}_{[m_s, \infty)}(n)}{k! \binom{n}{k}} \propto \frac{\mathbf{1}_{[m_s, \infty)}(n)}{\binom{n}{k}}$$

Using these likelihoods, we can compute the posterior in any given example.

By the above,

$$p(m | n) = \frac{|\Omega_{n,k,m}|}{|\Omega_{n,k}|} = \frac{\binom{m-1}{k-1}}{\binom{n}{k}}.$$

Therefore,

$$\mathbb{E}[m | n] = \binom{n}{k}^{-1} \sum_{m=k}^n m \binom{m-1}{k-1}.$$

But

$$m \binom{m-1}{k-1} = k \binom{m}{k},$$

so

$$\mathbb{E}[m \mid n] = k \binom{n}{k}^{-1} \sum_{m=k}^n \binom{m}{k}$$

But

$$\sum_{m=k}^n \binom{m}{k} = \binom{n+1}{k+1},$$

so

$$\mathbb{E}[m \mid n] = k \binom{n}{k}^{-1} \binom{n+1}{k+1}$$

But

$$\binom{n+1}{k+1} = \frac{n+1}{k+1} \binom{n}{k},$$

so

$$\mathbb{E}[m \mid n] = \frac{k}{k+1} (n+1)$$

Therefore,

$$\mathbb{E} \left[\left(\frac{k+1}{k} \right) m - 1 \right] = n.$$

11. GAMBLER'S RUIN

Suppose the player wins with probability p and the house wins with probability $1 - p$.

Let a_i be the probability that the fortune of a player with i dollars will hit k dollars before hitting 0 dollars. Note that $a_0 = 0$ and that $a_k = 1$.

We have

$$a_i = pa_{i+1} + (1-p)a_{i-1}$$

$$a_{i+1} = \frac{1}{p}a_i - \frac{1-p}{p}a_{i-1}$$

$$a_2 = \frac{1}{p}a_1$$

$$a_3 = \frac{1}{p}a_2 - \frac{1-p}{p}a_1$$

$$x^2 - \frac{1}{p}x + \frac{1-p}{p}$$

$$(x-1) \left(x - \frac{1-p}{p} \right)$$

$$a_i = A + B \left(\frac{1-p}{p} \right)^i$$

$$A + B = 0, \quad A + B \left(\frac{1-p}{p} \right)^k = 1$$

$$B \left(\left(\frac{1-p}{p} \right)^k - 1 \right) = 1$$

$$B = \frac{1}{\left(\frac{1-p}{p} \right)^k - 1}$$

$$a_i = \frac{\left(\frac{1-p}{p} \right)^i - 1}{\left(\frac{1-p}{p} \right)^k - 1}$$

12. ORDER STATISTICS

What is the distribution of $\max\{X_1, X_2\}$, where X_1 and X_2 are independent draws from the uniform distribution on $\{1, \dots, k\}$?

Let $Y = \max\{X_1, X_2\}$

$$\begin{aligned} \mathbb{P}(Y = i) &= \mathbb{P}(X_1 = X_2 = i) + \mathbb{P}(X_1 = i, X_2 < i) + \mathbb{P}(X_1 < i, X_2 = i) \\ &= \mathbb{P}(X_1 = i) \mathbb{P}(X_2 = i) + \mathbb{P}(X_1 = i) \mathbb{P}(X_2 < i) + \mathbb{P}(X_1 < i) \mathbb{P}(X_2 = i) \\ &= \frac{1}{k^2} + \frac{1}{k} \left(\frac{i-1}{k} \right) + \left(\frac{i-1}{k} \right) \frac{1}{k} \\ &= \frac{1}{k^2} (1 + 2(i-1)) \\ &= \frac{1}{k^2} (2i-1) \end{aligned}$$

$$\sum_{i=1}^k (2i-1) = 2 \sum_{i=1}^k i - k = 2 \frac{k(k+1)}{2} - k = k^2$$

So

$$\sum_{i=1}^k \mathbb{P}(Y = i) = 1,$$

as required, giving a check on our calculation.

13. BETA DISTRIBUTION

The beta distribution with parameters $a > 0$ and $b > 0$ has mean and variance given by

$$\mu = \frac{a}{a+b} = 1 - \frac{b}{a+b}, \quad \sigma^2 = \frac{ab}{(a+b)^2(a+b+1)}$$

We want to solve for a and b in terms of μ and σ .

Set

$$k = a + b.$$

Then $k > 0$.

Then

$$a = \mu k, \quad b = (1 - \mu)k.$$

In particular,

$$B(\mu k, (1 - \mu)k), \quad k > 0$$

is the family of beta distributions with mean μ .

Substitute:

$$\sigma^2 = \frac{\mu k(1 - \mu)k}{k^2(k + 1)} = \frac{\mu(1 - \mu)}{k + 1}$$

Note that $\sigma^2 < \mu(1 - \mu)$ because $k > 0$.

Solve for k :

$$k = \frac{\mu(1 - \mu)}{\sigma^2} - 1$$

Therefore:

$$a = \mu k = \mu \left(\frac{\mu(1 - \mu)}{\sigma^2} - 1 \right), \quad b = (1 - \mu)k = (1 - \mu) \left(\frac{\mu(1 - \mu)}{\sigma^2} - 1 \right)$$

14. LOCATION-SCALE FAMILIES

Let X be a random variable with distribution F . If u and v are real numbers with $v > 0$, write $F_{u,v}$ for the distribution function of the random variable $u + vX$. Since

$$u + vX \leq x \iff X \leq \frac{x - u}{v},$$

we have

$$F_{u,v}(x) = F\left(\frac{x - u}{v}\right).$$

If X has density f , write $f_{u,v}$ for the density of $u + vX$, then

$$f_{u,v}(x) = F'_{u,v}(x) = \frac{1}{v} F' \left(\frac{x-u}{v} \right) = \frac{1}{v} f \left(\frac{x-u}{v} \right)$$

15. MAXIMUM LIKELIHOOD ESTIMATION

$x_1, \dots, x_n \sim \text{Exp}(\lambda)$

$$\mathbb{P}(X > a) = \int_a^\infty \lambda e^{-\lambda x} dx = (-e^{-\lambda x})_a^\infty = e^{-\lambda a}$$

$$\begin{aligned} \mathbb{E}(x \mid x > a) &= \frac{1}{\mathbb{P}(X > a)} \int_a^\infty x \lambda e^{-\lambda x} dx \\ &= \int_a^\infty \lambda x e^{-\lambda(x-a)} dx \\ &= \int_0^\infty \lambda(y+a) e^{-\lambda y} dy \\ &= \int_0^\infty y \lambda e^{-\lambda y} dy + a \int_0^\infty \lambda e^{-\lambda y} dy \\ &= \mathbb{E}(X) + a \\ &= \frac{1}{\lambda} + a \end{aligned}$$

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim f(\mathbf{x}|\lambda')}(\log f(\mathbf{x} \mid \lambda) \mid \mathbf{x} > \mathbf{a}) &= \sum_{i=1}^n (\log \lambda - \lambda \mathbb{E}_{\mathbf{x} \sim f(\mathbf{x}|\lambda')}(x_i \mid x_i > a_i)) \\ &= n \log \lambda - \lambda \sum_{i=1}^n \left(\frac{1}{\lambda'} + a_i \right) \end{aligned}$$

16. HYPOTHESIS TESTING

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ be a random sample with μ and σ unknown.

We want to test the null hypothesis $\mu = \mu_0$ using the test statistic

$$T = \frac{\bar{X}_n - \mu_0}{S_n},$$

where

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Say we reject H_0 if $|T| > c$.

Under H_0 , $T \sim t_{n-1}$.

The level of the test is

$$\alpha = \mathbb{P}(|T| > c \mid H_0) = 2\mathbb{P}(T > c) = 2(1 - \mathbb{P}(T \leq c)) = 2(1 - \text{cdf}_{t_{n-1}}(c)).$$

We can choose c such that the test has level α by solving

$$\alpha = 2(1 - \text{cdf}_{t_{n-1}}(c))$$

for c in terms of α , giving

$$c = \text{ppf}_{t_{n-1}}(1 - \alpha/2).$$

17. POISSON DISTRIBUTION

The Poisson distribution with parameter λ is the discrete distribution with mass function

$$\mathbb{P}(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, \dots$$

$$\begin{aligned} \mathbb{E}(X) &= \sum_{x=0}^{\infty} x \mathbb{P}(X = x) \\ &= \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} \\ &= \lambda \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} \\ &= \lambda \end{aligned}$$

$$\begin{aligned} \mathbb{E}(X^2) &= \sum_{x=0}^{\infty} x^2 \mathbb{P}(X = x) \\ &= \lambda \sum_{x=1}^{\infty} \frac{x \lambda^{x-1}}{(x-1)!} e^{-\lambda} \\ &= \lambda^2 + \lambda \end{aligned}$$

$$\mathbb{V}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$f(\lambda) = \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} = \lambda e^{\lambda}$$

$$f'(\lambda) = \sum_{x=1}^{\infty} \frac{x \lambda^{x-1}}{(x-1)!} = (1 + \lambda) e^{\lambda}$$

$$\log f(x_1, \dots, x_n \mid \lambda) = \sum_i (x_i \log \lambda - \log x_i! - \lambda)$$

$$\frac{d}{d\lambda} f(x_1, \dots, x_n \mid \lambda) = \sum_i \left(\frac{x_i}{\lambda} - 1 \right) = \frac{x_1 + \dots + x_n}{\lambda} - n$$

$$\frac{d}{d\lambda} f(x_1, \dots, x_n \mid \lambda) = 0 \iff \lambda = \frac{x_1 + \dots + x_n}{n}$$

$$\hat{\lambda} = \frac{X_1 + \dots + X_n}{n}$$

$$\mathbb{E}[\hat{\lambda}] = \lambda, \quad \mathbb{V}(\hat{\lambda}) = \frac{\lambda}{n}$$

If n is large, then $\hat{\lambda}$ is approximately $N(\lambda, \lambda/n)$ -distributed.

In 1000 tosses of a coin, 560 heads and 440 tails appear. Is it reasonable to assume that the coin is fair? Justify your answer.

Suppose $X \sim \text{Binomial}(1000, 0.5)$. Then

$$\mathbb{P}(X \geq 560) \approx 8.25 \times 10^{-5}$$